

MSC Timer App

Overview

The MSC Timer App is a centralized solution designed to track PNR-based activities using Office ID and Queue selection. It enables users to start, pause, resume, and complete tasks while automatically capturing accurate working duration and ensuring consistent, structured data logging through a SharePoint-based system.

Business Context

The earlier Excel-based tracking system was effective but had inherent limitations due to its file-based nature, including restricted concurrent access and dependency on manual updates.

To overcome these challenges, a PowerApps-based solution has been developed to provide a scalable, real-time, and system-driven approach for activity tracking and monitoring.

PowerApps Solution – Key Benefits

1. Real-Time Multi-User Capability

- Multiple users can work simultaneously without conflicts
 - Eliminates Excel file locking and write delays
 - Fully supports concurrent operations
 - Successfully tested and validated for real-time usage
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2. Smart Time Tracking with Pause/Resume (New Enhancement)

- Users can **Start, Pause, Resume, and End** tasks
 - System captures only **actual working time** (pause time excluded automatically)
 - Ensures accurate effort tracking even during interruptions
 - Automatically saves progress during pause to prevent data loss
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3. Automated and Accurate Duration Calculation

- Start time captured instantly at task initiation
- End time recorded on completion

- Duration calculated automatically without manual input
 - Eliminates calculation errors and improves consistency
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4. Intelligent Workflow & Session Control

- Only **one active task per user** allowed at any time
 - Prevents multiple device usage for the same user
 - Ensures disciplined and controlled workflow execution
 - Covers both Loader and QC activities
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5. Advanced Validation & Business Rules

- PNR validation enforced (minimum 6 alphanumeric characters)
 - Special handling for specific offices (e.g., Shift Handling automation)
 - Automatic email subject cleaning and PNR extraction
 - Prevents invalid, duplicate, or incomplete data entry
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6. Controlled QC Process (Smart & Scalable)

- Supports up to **3 Quality Check levels (QC1, QC2, QC3)**
 - QC automatically attaches to the relevant PNR + Queue record
 - QC cannot start before loader completion
 - Prevents unnecessary QC levels — only used when required
 - Ensures process clarity and avoids duplication
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7. Pause Visibility & Smart Locking (New Enhancement)

- Tasks on **Pause are clearly identified and protected**
 - Only the same user who paused the task can resume it
 - Prevents other users from modifying paused tasks
 - Ensures data integrity and avoids conflicts
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8. Centralized Data & Visibility

- All data stored in SharePoint for centralized access
- Provides real-time visibility across teams

- Enables reporting, monitoring, and audit readiness
- Maintains a single source of truth

9. Strong Data Governance

- Users can only **create and complete tasks**
- No direct editing or deletion allowed
- Ensures immutability and audit compliance
- Prevents accidental or unauthorized data changes

Key Improvements over Excel

Area	Excel System	PowerApps Solution
Data Entry	Manual controlled +	Fully automated
Concurrency	Limited (file-based)	Verified real-time multi-user
Time Tracking	System Based	Start–Pause–Resume–End
Validation	Static	Dynamic and rule-based
Workflow Control	Limited	Controlled and enforced
QC Handling	Manual	Structured & scalable
Data Integrity	Editable	Controlled & secure
Scalability	Limited	Scalable & future-ready

Testing & Validation Scope

The solution has been tested and validated across:

- Multi-user concurrent usage
- Single-session enforcement (multi-device prevention)
- Pause/Resume functionality with accurate time tracking
- QC multi-level handling (QC1 → QC3)
- Duplicate and edge-case scenarios
- Data consistency and integrity in SharePoint

Conclusion

The MSC Timer App successfully transforms the process from a manual, file-based system into a fully automated, scalable, and controlled platform.

With enhancements such as **Pause/Resume tracking, intelligent workflow control, session locking, and structured QC handling**, the solution ensures accurate time capture, operational efficiency, and strong data governance.

The application has been validated under real-world conditions and is now **production-ready**, supporting reliable, conflict-free operations across teams.

The screenshot displays the MSC Timer App interface. At the top, there is a header bar with the Power Apps logo and the text "Power Apps | MSC_Timer_APP". Below the header, there is a green banner with the CATHAY logo and the text "MSC_Queue_Log_DWNL". The main form contains three input fields: "Office_ID" with a dropdown menu showing "BOMCX0431", "Queue" with a dropdown menu showing "Q12C0", and "PNR/Subject" with a text input field containing "mnop12". Below the input fields, there are three buttons: "Start Timer" (green), "End Timer" (grey), and "Pause Task" (grey).